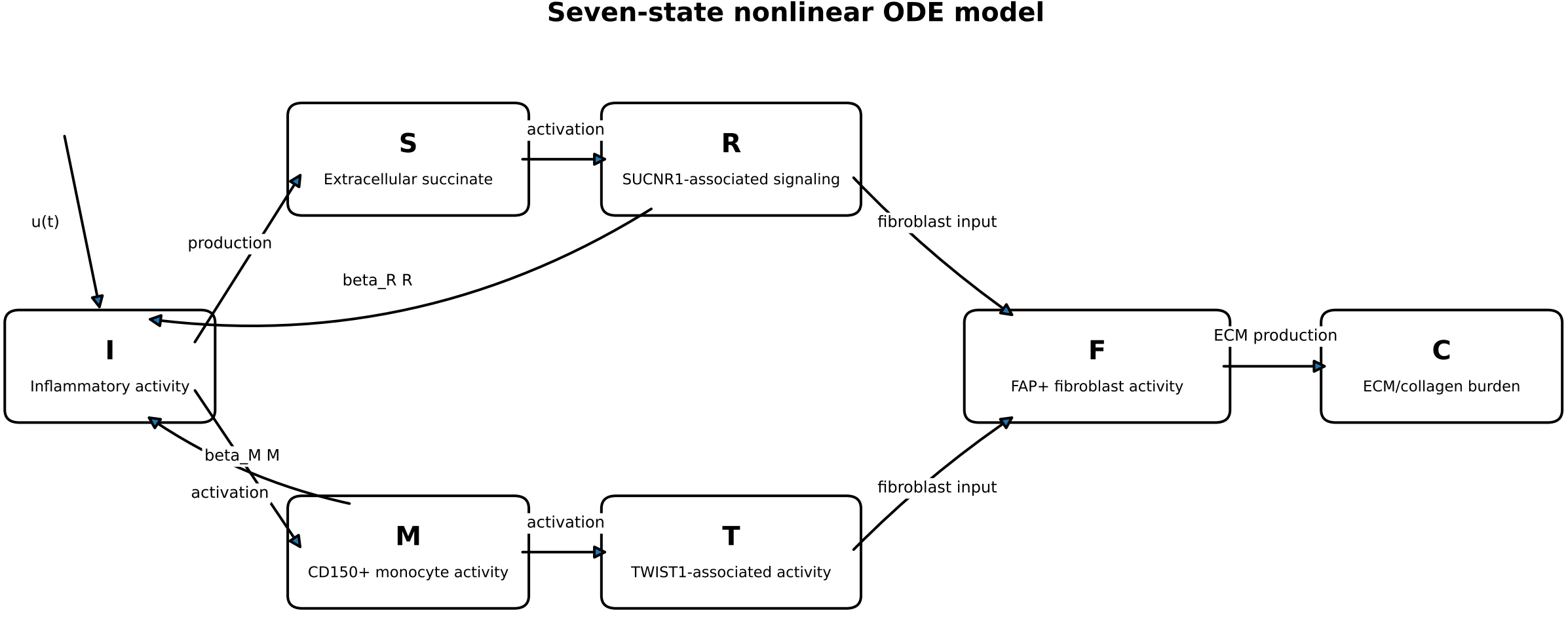
Seven-state nonlinear ODE model
S
Extracellular succinate
activation
R
SUCNR1-associated signaling
u(t)
production
beta_R R
fibroblast input
I
Inflammatory activity
F
FAP+ fibroblast activity
ECM production
C
ECM/collagen burden
beta_M M
activation
fibroblast input
M
CD150+ monocyte activity
activation
T
TWIST1-associated activity
Two upstream signaling routes converge additively at F; no direct R–T interaction is assumed.